SOLAR ENERGY PROPOSAL · ข้อเสนอพลังงานแสงอาทิตย์

Solar Power for TM Concrete Factory

110.8 kWp unified rooftop solar system across 4 industrial buildings — Koh Phangan, Surat Thani

ระบบโซลาร์เซลล์ 110.8 kWp บนหลังคา 4 อาคารโรงงาน — เกาะพะงัน สุราษฎร์ธานี

฿835,543	3.8 yrs	139,257	25 yrs
Annual Savings / ประหยัดต่อปี	Payback Period / คืนทุน	kWh/year Production	Panel Warranty

Prepared for	Proposal Date	Reference	Valid Until
TM Concrete Factory	**14 March 2026**	**TM-2026-004**	**13 April 2026 (30 days)**

YOUR PROJECT · โปรเจกต์ของคุณ

Four Buildings, One Central System

191 solar panels across 4 industrial buildings, connected to one central 100kW Huawei inverter — maximum efficiency, minimal infrastructure.

แผงโซลาร์ 191 แผงบน 4 อาคาร เชื่อมต่อกับอินเวอร์เตอร์ส่วนกลาง 100kW — ประสิทธิภาพสูงสุด

Aerial Rendering — 191 Panels · 110.8 kWp across 4 Buildings

System Layout — Central Inverter · DC Cable Routes

🏭 PROPERTY DETAILS / รายละเอียดโครงการ

Property	**TM Concrete Factory**
Location	**Koh Phangan, Surat Thani**
Buildings	**4 industrial buildings**
System Type	**Unified — 1 central inverter**

☀️ ROOF ASSESSMENT / ประเมินหลังคา

Roof Type	**Gable — Metal Sheet + Iscorit**
Total Usable	**~620 m² (south-facing slopes)**
Roof Pitch	**~13° (from architectural plans)**
Shading	**None — fully exposed ✓**

🏭

4 Buildings

A · B · C · Iscorit

⚡

1 Inverter

Central SUN2000-100KTL

☀️

110.8 kWp

191 LONGi 580W panels

💰

฿836K/yr

Annual savings

BUILDING BREAKDOWN · รายละเอียดแต่ละอาคาร

Four Roofs, One Unified System

All panels feed into a single central inverter positioned between the buildings for minimal DC cable runs.

แผงทั้งหมดเชื่อมกับอินเวอร์เตอร์กลาง 1 ตัว เพื่อลดความยาวสาย DC

Building	Dimensions	Panels	kWp	Layout	MPPT	Annual Savings
🏗️ **Building A**	15.9 × 8.5m	**28**	16.2	2r × 14c portrait	MPPT 1	**฿122,488**
🏗️ **Building B**	15.9 × 8.5m	**28**	16.2	2r × 14c portrait	MPPT 2	**฿122,488**
🏗️ **Building C**	17.3 × 10.5m	**35**	20.3	5r × 7c landscape	MPPT 3	**฿153,110**
🏭 **Iscorit Building**	~29 × 17m	**100**	58.0	4r × 25c portrait	MPPT 4-6	**฿437,457**
TOTAL		**191**	**110.8**		6 MPPT	**฿835,543**

CHOOSE YOUR PLAN · เลือกแผนของคุณ

Three Options, One Goal: Lower Bills

All options use the same premium equipment. Choose what fits your budget.

ทุกตัวเลือกใช้อุปกรณ์คุณภาพสูงเหมือนกัน เลือกตามงบประมาณ

Option A: PPA

Zero upfront — we install, you save / ไม่ต้องจ่ายล่วงหน้า

฿0 Upfront

pay only ฿4.50/kWh (25% less than PEA)

What is PPA? We install solar at zero cost. You buy electricity at ฿4.50/kWh — 25% cheaper than PEA. เราติดตั้งโซลาร์ฟรี คุณเพียงซื้อไฟฟ้าในราคา ฿4.50/kWh — ถูกกว่า PEA 25%

- ฿0 installation cost
- 25% cheaper than PEA rate
- Free maintenance 20 years
- Free panel cleaning 2×/year
- System insured
- Start saving Day 1

Monthly Savings vs PEA

~฿17,400/mo

★ RECOMMENDED / แนะนำ

Option B: EPC

Own your system outright / เป็นเจ้าของระบบ

฿3,212,620

one-time payment (฿29/Wp)

Payment Terms: 30% deposit · 40% on delivery · 30% on commissioning

- 191 panels + 100kW inverter — fully installed
- PEA license & grid connection
- 25-year panel warranty
- 10-year inverter warranty
- 24/7 monitoring (1yr free)
- System insured
- You own it — all savings are yours

Annual Savings

฿835,543

Option C: EPC + Battery

Solar + emergency backup / โซลาร์ + สำรองไฟฉุกเฉิน

฿3,772,620

solar + 40 kWh battery storage

Payment Terms: 30% deposit · 40% on delivery · 30% on commissioning

- Everything in Option B, plus:
- 2× Huawei LUNA2000-20kWh = 40 kWh
- Critical load backup during blackouts
- Office + security + control systems protected
- 10-year battery warranty
- Smart energy management via app

Annual Savings

฿835,543

฿208,888/yr saved vs PEA rate	Payback: **3.8 years** / คืนทุน 3.8 ปี	Payback: **4.5 years** · + blackout protection

SIDE-BY-SIDE COMPARISON · เปรียบเทียบ

Compare All Three Options

	☀️ PPA	⚡ EPC ★	🔋 EPC+Battery
Upfront Cost	**฿0**	**฿3,212,620**	**฿3,772,620**
Monthly Payment	~฿52,200/mo ฿4.50 × kWh	฿0 after purchase	฿0 after purchase
Annual Savings	**~฿209K**	**฿835,543**	**฿835,543**
Payback Period	Immediate	**3.8 years**	**4.5 years**
Blackout Protection	❌ Daytime only	❌ Daytime only	✅ **Critical loads**
Ownership	TM Energy (buy after 20 yrs)	You own it ✓	You own it ✓

SYSTEM DETAILS · รายละเอียดระบบ

Industrial-Grade Equipment

☀️ Solar Panels — LONGi Hi-MO X6

Model	LR5-72HGD 580W
Quantity	191 panels
Total Capacity	110.8 kWp
Technology	N-type TOPCon · 22.3% eff.
Warranty	25-year product · 30-year perf.

🔋 Battery (Option C) — LUNA2000

Configuration	2× LUNA2000-20kWh = 40 kWh
Output Power	10 kW (continuous)
Purpose	Emergency backup — critical loads
Cost	฿560,000
Warranty	10 years · 6,000+ cycles

⚡ Central Inverter — Huawei SUN2000-100KTL

Quantity	1 unit — 100 kW
MPPT	6 trackers (one per building string)
DC:AC Ratio	1.11:1 (optimal)
Efficiency	98.8% · IP66 outdoor
Warranty	10 years

🔧 Installation Includes

Mounting	No-drill roof clamps
Wind Rating	Up to 60 m/s (typhoon rated)
DC Cabling	Max 25m runs to central inverter
Protection	Surge + breakers + grounding
PEA License	Application handled by us

YOUR SAVINGS · เงินที่ประหยัด

Massive Industrial Savings

฿835,543	฿8.4M	3.8	69.6t
Annual Savings	10-Year Savings	Years to Payback	CO_2 Saved/Year

TIMELINE · ขั้นตอน

From Approval to Power in 8–10 Weeks

WEEK 1-2

Survey & Design

Site measurement, structural assessment, system design

WEEK 3-4

Permits & Procurement

PEA license, equipment order (100kW inverter lead time)

WEEK 5-8

Installation

191 panels across 4 buildings, central inverter, DC cabling

WEEK 9-10

Go Live ✓

Testing, PEA inspection, grid connection, monitoring

WARRANTIES · การรับประกัน

Protected for Decades

☀️

25 years

Panel Warranty

LONGi · 30-year performance

⚡

10 years

Inverter Warranty

Huawei SUN2000-100KTL

🔋

10 years

Battery Warranty

LUNA2000 · 6,000+ cycles

🔧

24 months

Workmanship

TM Energy installation

📡

24/7

Monitoring

FusionSolar App

🏝️

Local

On-Island Team

Same-day response

Ready to Power Your Factory with Solar?

This proposal is valid for 30 days.

💬 **WhatsApp Us Now** 📞 **062-348-0119**

1. Choose your option **2.** Sign agreement **3.** We start within 1 week

Service Agreement

SELECTED OPTION

- ○ Option A: PPA — ฿0 Upfront
- ○ Option B: EPC — ฿3,212,620
- ○ Option C: EPC+Battery — ฿3,772,620

KEY TERMS

- System: 110.8 kWp across 4 buildings (191× LONGi 580W panels)
- Central inverter: 1× Huawei SUN2000-100KTL-M3 (100 kW)
- If Option C: + 2× Huawei LUNA2000-20kWh (40 kWh total)
- Installation: 8–10 weeks from signing
- Warranty: 25-year panels, 10-year inverter, 10-year battery, 24-month workmanship

▸ **Terms & Conditions**

CLIENT	TM ENERGY
_____	_____
FULL NAME	FULL NAME
DATE	DATE

ABOUT US

About TM Energy

TM Energy delivers professional solar solutions across Koh Phangan. Design, install, and maintain high-performance solar systems — with a local team providing same-day service.

Powered by TM Developers

TM ENERGY

Powered by TM Developers — Koh Phangan

📞 062-348-0119 💬 wa.me/66623480119 📍 Koh Phangan

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